Resume
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The year 2023 marks a turning point for researchers and learners. Generative AI has changed my life. It's just like how the X-ray was invented in 1895, which opened a door to many discoveries later in the 20th century. Now, people who aren't smart can also do great work. where a discovery mindset and motivation for innovation are prioritized over technical skills more than ever before. I have been following the development in the GenAI field very closely. My work will be AI-integrated.

Research/Work

- **Research Assistant for Precision Swine Production - University of Missouri 07/2024 - 12/2024**
- Develped solutions to predict estrous time in swine: configured MCU, and LIDAR/IR cameras with codes to become an inspection robot; trained in models for imaging segmentation and transformer neural network. Filled an NSF fellowship proposal; a two-page extended abstract proposed that the ground truth uncertainty issue might be addressed by using a microneedle patch for continuous estradiol monitoring
- **Bioinformatics Research Assistant - Florida Atlantic University/Texas Tech University, 03/2023 - 06/2023, 04/2024 - 06/2024**
- Performed mutation and transposable element between samples from whole exome/genome data, using GATK/BWA/Samtools/Bcftools/Pysam/Snpeff/customized python code, short-term contractor working with the same team.
- **Research Data Assistant - IU School of Medicine, 06/2021 - 07/2022**
- Analyzed over 50 various RNA-seq projects at the sequencing core, including: processing raw RNA sequencing data, mapping, sample outlier detection, batch effect removal, various DGE analysis, pathway analysis, splicing analysis, and troubleshooting
- Generated html report with the newest visualization tools/plots for the users
- Processed raw WES data, SNP, INDEL analysis
- Developed a GUI database
- Provided consultation to researchers regarding the analysis results
- Mentored undergraduate and master's students

Education

- **Arizona State University, 01/2025 – present, M.S. Electrical Engineering, online** .
  Took VLSI design, Digital Signal Processing (DSP) courses

- **San Diego State University, 01/2025 – present, M.S. Medical Physics, onsite**
  Courses in medical physics, clinical lab in diagnostic imaging and radiation therapy, CAMPEP program. Shadowed in a MEMS lab with research using electrochemistry and graphene to monitor glucose levels in real time. Research in blockchain data systems for agriculture and medical imaging applications. Teaching Associate for the electricity lab. Selected for two fellowships in research. Expected all courses to be completed in Fall 2025. Took upper-level physics courses equivalent to physics minor prior enrolling.

- **University of Missouri Columbia, 08/2024-12/2024, master student** .
  Took courses in VHDL digital logic devices, CMOS, Virtuoso, Vivado

- **Purdue University, 08/2023-05/2024, non-degree student taking medical physics courses,**
  Took courses about how imaging begging generated through X-ray, CT, PET, MR, or ultrasound; conducted MR research that isolated non-water proton signals, led in a poster presentation at the school conference; developed TG51 online calculator for radiation dosage calculation led to oral presenting in *regional* conference

2017-2021

- **Indiana University – Purdue University Indianapolis (IUPUI), B.S. in Biomedical Informatics,** 150 credits in liDfe science, statistics, computer science, bioinformatics, Internship to identify potential pathways and markers related to mitochondria in RGC cells from single-cell sequencing data.

- **Saint Louis Community College, University of Nebraska Omaha, Ivy Tech Community College**
  NSF-funded 16 credits of hands-on training in biotechnology, Indiana General Education Transfer Core Certificate

Projects

- **Genomics Bioinformatics (2020-2023): Whole genome SNP/INDEL calling, transposable-element mapping, large-scale RNA/miRNA/miRNA/lncRNA pipelines, multi-generation family inheritance mutation calling, possible biological meaning with literature search; single cell functional check** https://ax3301.github.io/RGC/rgc_2021.html**; dashboard development** https://ax3301.github.io/PDAC_2022
- **Web Scraping + Functional call: Toolkit for new medical physics students** https://ax3301.github.io/Medical_Physics_Toolkit_23/
- **NSF GRFP Proposal 2023: Large language models for advancements in biomedical research** https://ax3301.github.io/RGC/NSF_2023.pdf
- **NSF GRFP Proposal 2024: Precision livestock farming using engineering strategies in swine Barns** https://ax3301.github.io/RGC/NSF_2024.pdf
- **Water signal suppression in NMR imaging** https://github.com/ax3301/RGC/blob/main/NMR_2024.ipynb
- **Swine Vision: Imaging segmentation and time series transformer neural network in swine** https://github.com/ax3301/Pig_Vision_2024
- **NSF New Grants Tracker and AI Agent** https://nsf-grant-tracker-2025.sequencingmaster.com/ https://nsf-grant-tracker-2025.sequencingmaster.com/agent
- **California State University Agriculture Research Institution Student Fellowship Proposal 2025: Blockchain-enabled agricultural data system with AI for swine and soybean production (Selected for 10k)** https://ax3301.github.io/RGC/F_2025.pdf
- **Developing Blockchain Workflow for Quality Control in Medical Physics_2025** ttps://docs.google.com/presentation/d/1UhgE93MdXfVbx7gheiY_hSyWWwMQ-oUh/edit?usp=sharing&ouid=111482565369766842097&rtpof=true&sd=true https://imaging-mp-record.pages.dev/

Publication, Poster, Oral Presentation

- Chang H. (2021). A comprehensive list of undergraduate bioinformatics programs in the United States. https://doi.org/10.31219/osf.io/9g5wu
- Qiuhong He, et al. In Vivo pi-SSelMQC to Recover 100% Biomarker Signals From Both MQ Coherence Transfer Pathways with Applications in Tumor Lactate Imaging. Academic Radiology. https://doi.org/10.1016/j.acra.2025.11.042, https://ax3301.github.io/RGC/HSCI_032024.pdf
- Chang H. (Apr 2024). The purpose of developing an online TG51 calculator for quick beam measurement: To provide a convenient and accessible tool for trainees. Oral presentation at: American Association of Physicists in Medicine ORVC 2024 Spring Meeting. https://ax3301.github.io/RGC/ORVC_2024.pdf

Technical Skills

Languages & Tools: Python, R, MATLAB, VHDL, SystemVerilog, HTM, Bash
Hardware & EDA: Microcontrollers, LiDAR/IR sensors, Virtuoso, Vivado
AI & ML: PyTorch, Transformers, OpenCV, Prompt Engineering
Bioinformatics: GATK, BWA, SAMtools, BCFtools, SnpEff, Pysam, Seurat, DESeq2
Data & Visualization: Pandas, Plotly, Dash, Matplotlib, ggplot2